BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 2: Writing Code from a Scenario

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What must be true before binary search is used?

2. In OCR ERL, what is 2 ^ 3?

A) 6 B) 8
C) 9 D) 23

3. "OR is false whenever both inputs are true." Correct this. Explain why.

4. Join text '12' to '3'. Then convert '12' to an integer and add 3. Give both results. What should happen if the input is 'twelve'?

Today's objectives

Explain the core ideas of paper 2: writing code from a scenario using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Identify input

List all inputs, types and relevant constraints before writing code.

Design

State algorithm steps separately from program syntax.

Select and iterate

Write appropriate IF/ELSE and loops rather than unnecessary repetition.

Output and test

Check both a typical value and at least one boundary/unexpected condition.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Identify input	List all inputs, types and relevant constraints before writing code.
Design	State algorithm steps separately from program syntax.
Select and iterate	Write appropriate IF/ELSE and loops rather than unnecessary repetition.
Output and test	Check both a typical value and at least one boundary/unexpected condition.

I Do - One worked example

Write an algorithm to input three integers and output their total.

1. Set total=0.
2. Repeat three times to input a number.
3. Add each entered number to total.
4. Output total after the loop.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Design a program to input five marks and count those >= 50.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Cast the typed file size to an integer before comparing it with 100.

```
sizeText = input("File size in MB?")
sizeMB = ...
if sizeMB > 100 then
print("Large")
else
print("Small")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Write OCR-style pseudocode to output the larger of two input values. [4 marks]

Q3. Explain why an accumulator starts at zero when summing values. [2 marks]

Q4. State why a final output statement should be outside a loop when only one total is required. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. The same program line must be copied five times instead of using a loop. ☐ T ☐ F
2. List all inputs, types and relevant constraints before writing code. ☐ T ☐ F
3. State algorithm steps separately from program syntax. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 The list must be sorted. This lets the search safely rule out half the remaining items. ☐

Q2 B - 8 ☐

Q3 When at least one input is true, including when both are true. ☐

Q4 Joining the strings gives '123'. Converting '12' to an integer and adding 3 gives 15. Handle or reject 'twelve': it cannot be cast to an integer directly. ☐

Code First - Model

```
sizeText = input("File size in MB?")
sizeMB = int(sizeText)
if sizeMB > 100 then
print("Large")
else
print("Small")
endif
```

Output: Large (if input is "120")

We Do - Check the model

1. Print passCount after all entries. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Input a,b; if a>b print(a), else print(b); equality may output either as both are largest. ☐

Q3 Zero is the additive identity and avoids introducing an unwanted starting total. ☐

Q4 Inside the loop would print intermediate totals repeatedly. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	**Check my understanding**